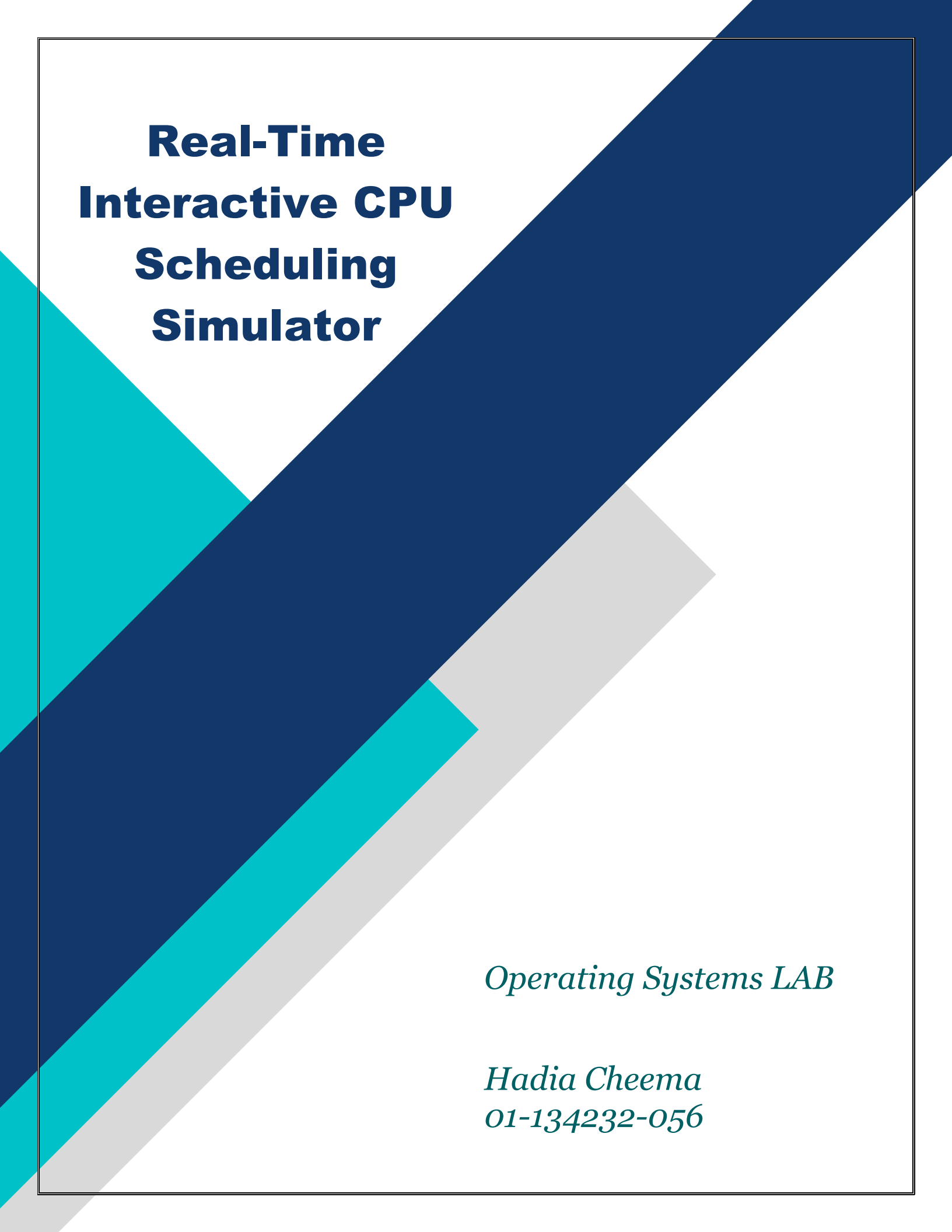




Real-Time Interactive CPU Scheduling Simulator

Operating Systems LAB

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INTRODUCTION:

CPU scheduling is one of the most important functions of an operating system. In a computer system, many processes may be waiting for CPU execution. The operating system uses scheduling algorithms to decide which process should get the CPU first. This project is a **Real-Time Interactive CPU Scheduling Simulator**. It helps users understand how different CPU scheduling algorithms work. The simulator provides a graphical user interface where users can add processes, select algorithms, run simulations, view Gantt charts, and analyze performance metrics. The project is useful for learning because it shows process execution visually instead of only giving console output.

Problem Statement:

The problem is to design and implement a GUI-based CPU scheduling simulator that can simulate multiple scheduling algorithms and compare their performance.

The simulator should allow the user to:

- Add processes with arrival time, burst time, and priority.
- Select different scheduling algorithms.
- View the execution order using a Gantt chart.
- Calculate performance metrics.
- Compare different algorithms.
- Get adaptive feedback based on results.

Objectives

The main objectives of this project are:

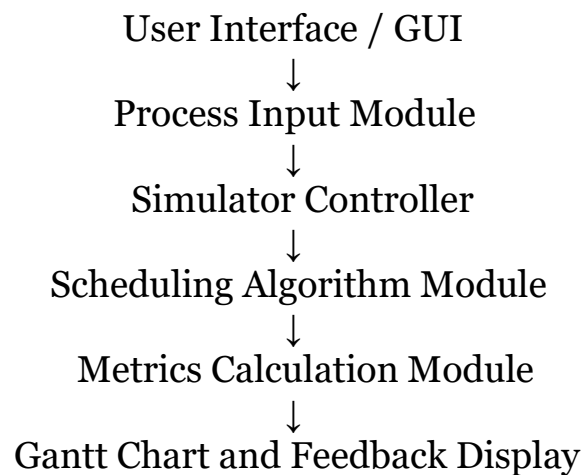
1. To implement multiple CPU scheduling algorithms.
2. To create an interactive GUI-based simulator.
3. To show process execution through a Gantt chart.
4. To calculate important performance metrics.
5. To compare algorithms based on performance.
6. To provide adaptive feedback to recommend better scheduling strategies.
7. To improve understanding of operating system scheduling concepts.

TOOLS USED:

TOOLS	PURPOSE
Python	Main programming language
PyQT6	GUI development
VS CODE	Code Editor
Pyinstaller	Creating executable file
Windows	Development envir

System Architecture

The system is divided into different modules to keep the project clean and easy to manage.



Scheduling Algorithms Implemented

First-Come, First-Served

FCFS executes processes in the order they arrive. The process that arrives first gets the CPU first. It is simple and easy to implement, but it can cause long waiting time.

Shortest Job First

SJF selects the process with the shortest burst time from the ready queue. It reduces average waiting time, but long processes may wait for a long time.

Shortest Remaining Time First

SRTF is the preemptive version of SJF. It always selects the process with the shortest remaining time. If a new process arrives with a smaller remaining time, the current process is interrupted.

Priority Scheduling

Priority scheduling selects the process based on priority. In this project, a smaller priority number means higher priority.

Two types are implemented:

- Priority Non-Preemptive
- Priority Preemptive

Round Robin

Round Robin gives each process a fixed time quantum. If the process does not finish within that time, it goes back to the ready queue. It is useful for time-sharing systems.

Multilevel Feedback Queue

MLFQ uses multiple queues with different priority levels and time quantum. New processes start in the highest priority queue. If they do not finish, they move to lower priority queues.

Performance Metrics

The simulator calculates the following metrics:

Average Waiting Time

Waiting time is the time a process waits before execution.

Waiting Time = Turnaround Time - Burst Time

Turnaround Time

Turnaround time is the total time from process arrival to completion.

Turnaround Time = Completion Time - Arrival Time

Response Time

Response time is the time from process arrival to first CPU execution.

Response Time = Start Time - Arrival Time

CPU Utilization

CPU utilization shows how much time the CPU was busy.

CPU Utilization = Busy Time / Total Time $\times$ 100

Throughput

Throughput shows how many processes are completed per unit time.

Throughput = Completed Processes / Total Time

GUI Features

The simulator has a modern dashboard-style GUI. It includes:

1. Process input form
2. Algorithm selection dropdown
3. Time quantum input
4. Run simulation button
5. Delete and clear process options
6. Live Gantt chart
7. Process table
8. Metrics panel
9. Adaptive feedback panel
10. Algorithm comparison button

The GUI makes the simulator easy to use and helps visualize how processes are executed

Sample Input

Process	ID Arrival	Time Burst	Time Priority
P1	1	4	1
P2	2	5	1
P3	5	10	4

Sample Output

For FCFS scheduling, the execution order is:
 IDLE → P1 → P2 → P3

Sample metrics:

Metric	Value
Average Waiting Time	2.67
Average Turnaround Time	9.0
Average Response Time	2.67
CPU Utilization	95.0%
Throughput	0.15

Testing

The following features were tested:

Feature	Status
Add Process	Passed
Delete Process	Passed
Clear All	Passed
FCFS	Passed
SJF	Passed
SRTF	Passed
Priority Scheduling	Passed
Round Robin	Passed
MLFQ	Passed

Feature

Gantt Chart
Metrics Calculation
Algorithm Comparison
Adaptive Feedback

Status

Passed
Passed
Passed
Passed

SCREENSHOTS:

CPU Scheduler

Simulation Dashboard

FCFS

SJF

SRTF

Priority Non-Preemptive

Priority Preemptive

Round Robin

MLFQ

Compare Algorithms

Real-Time Interactive CPU Scheduling Simulator

Process Configuration

Process ID

P1

Arrival Time

0

Burst Time

1

Priority

1

Algorithm

FCFS

Time Quantum

2

Add Process

Delete Selected

Clear All

Run Simulation

Performance Metrics

Average Waiting Time: -

Average Turnaround Time: -

Average Response Time: -

CPU Utilization: -

Throughput: -

Completion Order: -

Live Gantt Chart

Gantt Chart will appear here

Process Table

PID	Arrival	Burst	Remaining	Priority	Start	Completion	Waiting	Turnaround	Response	Status
-----	---------	-------	-----------	----------	-------	------------	---------	------------	----------	--------

Activate Windows
Go to Settings to activate Windows.

CPU Scheduler

Simulation Dashboard

FCFS

SJF

SRTF

Priority Non-Preemptive

Priority Preemptive

Round Robin

MLFQ

Compare Algorithms

Live Gantt Chart

Gantt Chart will appear here

Process Table

PID	Arrival	Burst	Remaining	Priority	Start	Completion	Waiting	Turnaround	Response	Status
-----	---------	-------	-----------	----------	-------	------------	---------	------------	----------	--------

Adaptive Feedback

Adaptive feedback will appear after running simulation.

Activate Windows
Go to Settings to activate Windows.

FCFS:

Real-Time Interactive CPU Scheduling Simulator

Process Configuration

Process ID

P1

Arrival Time

0

Burst Time

1

Priority

1

Algorithm

FCFS

Time Quantum

2

Add Process

Delete Selected

Clear All

Run Simulation

Performance Metrics

Average Waiting Time: 8.33

Average Turnaround Time: 15.0

Average Response Time: 8.33

CPU Utilization: 100.0

Throughput: 0.15

Completion Order: ['P1', 'P2', 'P3']

Live Gantt Chart

P1

P2

P3

0

12

15

20

Process Table

PID	Arrival	Burst	Remaining	Priority	Start	Completion	Waiting	Turnaround	Response	Status
P1	0	12	0	1	0	12	0	12	0	Completed
P2	0	3	0	1	12	15	12	15	12	Completed
P3	2	5	0	1	15	20	13	18	13	Completed

Activate Windows

Go to Settings to activate Windows.

SJF:

Process Configuration

Process ID

P1

Arrival Time

0

Burst Time

1

Priority

1

Algorithm

SJF

Time Quantum

2

Add Process

Delete Selected

Clear All

Run Simulation

Performance Metrics

Average Waiting Time: 3.0

Average Turnaround Time: 9.67

Average Response Time: 3.0

CPU Utilization: 100.0

Throughput: 0.15

Completion Order: ['P2', 'P3', 'P1']

Live Gantt Chart

P2

P3

P1

0

3

8

20

Process Table

PID	Arrival	Burst	Remaining	Priority	Start	Completion	Waiting	Turnaround	Response	Status
P1	0	12	0	1	8	20	8	20	8	Completed
P2	0	3	0	1	0	3	0	3	0	Completed
P3	2	5	0	1	3	8	1	6	1	Completed

Activate Windows

Go to Settings to activate Windows.

SRTF:

Process Configuration

Process ID

P1

Arrival Time

0

Burst Time

1

Priority

1

Algorithm

SRTF

Time Quantum

2

Add Process

Delete Selected

Clear All

Run Simulation

Performance Metrics

Average Waiting Time: 2.75

Average Turnaround Time: 8.0

Average Response Time: 2.5

CPU Utilization: 100.0

Throughput: 0.19

Completion Order: ['P2', 'P4', 'P3', 'P1']

Live Gantt Chart

P2

P3

P4

P3

P1

0

3

5

6

9

21

Process Table

PID	Arrival	Burst	Remaining	Priority	Start	Completion	Waiting	Turnaround	Response	Status
P1	0	12	0	1	9	21	9	21	9	Completed
P2	0	3	0	1	0	3	0	3	0	Completed
P3	2	5	0	1	3	9	2	7	2	Completed
P4	5	1	0	4	5	6	0	1	0	Completed

PRIORITY NON PREEMPTIVE:

Process Configuration

Process ID

P1

Arrival Time

0

Burst Time

1

Priority

1

Algorithm

SRTF

Time Quantum

2

Add Process

Delete Selected

Clear All

Run Simulation

Performance Metrics

Average Waiting Time: 2.25

Average Turnaround Time: 7.0

Average Response Time: 0.0

CPU Utilization: 100.0

Throughput: 0.21

Completion Order: ['P2', 'P4', 'P3', 'P1']

Live Gantt Chart

P1

P3

P2

P4

P3

P1

0

2

4

5

6

9

19

Process Table

PID	Arrival	Burst	Remaining	Priority	Start	Completion	Waiting	Turnaround	Response	Status
P1	0	12	0	1	0	19	7	19	0	Completed
P3	2	5	0	1	2	9	2	7	0	Completed
P4	5	1	0	4	5	6	0	1	0	Completed
P2	4	1	0	3	4	5	0	1	0	Completed

PRIORITY PREEMPTIVE:

Real-Time Interactive CPU Scheduling Simulator

Process Configuration

Process ID

P1

Arrival Time

0

Burst Time

1

Priority

1

Algorithm

Priority Preemptive

Time Quantum

2

Add Process

Delete Selected

Clear All

Run Simulation

Performance Metrics

Average Waiting Time: 9.0

Average Turnaround Time: 13.75

Average Response Time: 9.0

CPU Utilization: 100.0

Throughput: 0.21

Completion Order: ['P1', 'P3', 'P2', 'P4']

Live Gantt Chart

P1

P3

P2

P4

0

12

17

18

19

Process Table

PID	Arrival	Burst	Remaining	Priority	Start	Completion	Waiting	Turnaround	Response	Status
P1	0	12	0	1	0	12	0	12	0	Completed
P3	2	5	0	1	12	17	10	15	10	Completed
P4	5	1	0	4	18	19	13	14	Activate Windows Go to Settings to activate Windows.	Completed
P2	4	1	0	3	17	18	13	14	13	Completed

ROUNDROBIN:

Real-Time Interactive CPU Scheduling Simulator

Process Configuration

Process ID

P1

Arrival Time

0

Burst Time

1

Priority

1

Algorithm

Round Robin

Time Quantum

2

Add Process

Delete Selected

Clear All

Run Simulation

Performance Metrics

Average Waiting Time: 4.75

Average Turnaround Time: 9.5

Average Response Time: 1.5

CPU Utilization: 100.0

Throughput: 0.21

Completion Order: ['P2', 'P4', 'P3', 'P1']

Live Gantt Chart

P1

P3

P1

P2

P3

P4

P1

P3

P1

0

2

4

6

7

9

10

12

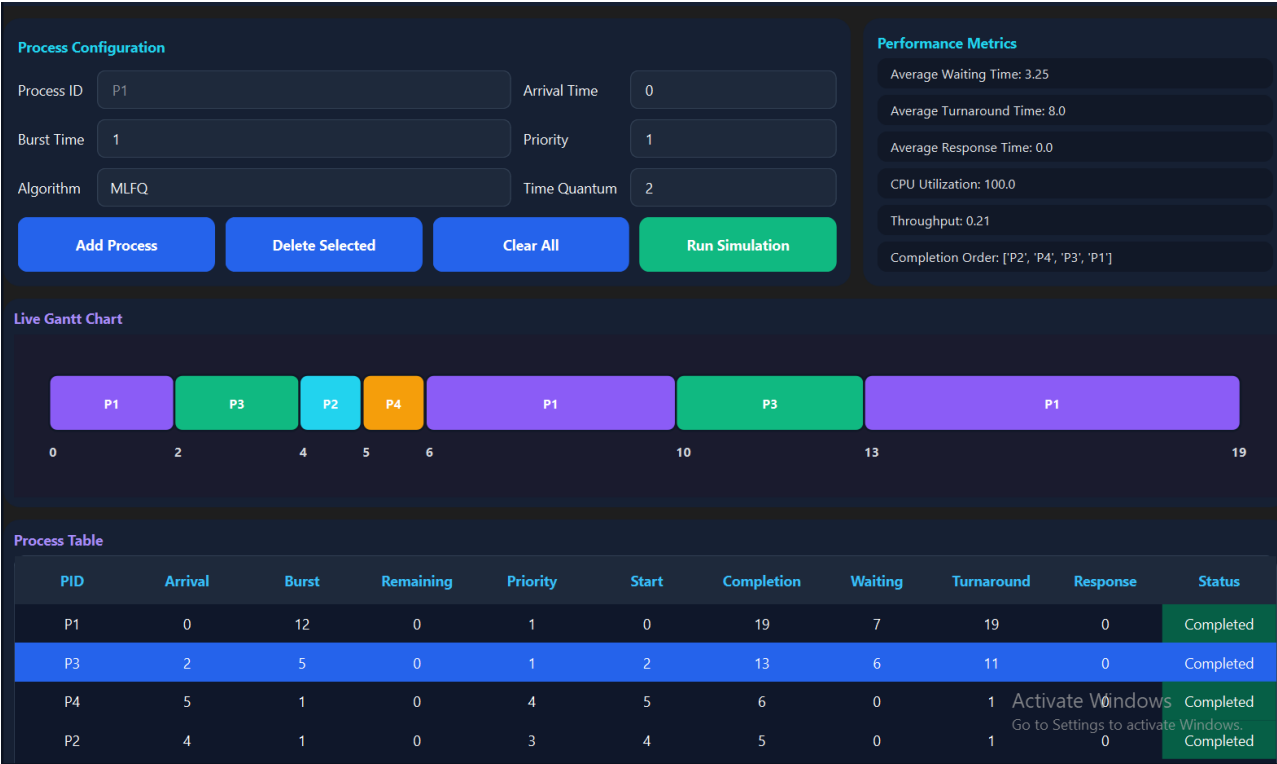
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19

Process Table

PID	Arrival	Burst	Remaining	Priority	Start	Completion	Waiting	Turnaround	Response	Status
P1	0	12	0	1	0	19	7	19	0	Completed
P3	2	5	0	1	2	13	6	11	0	Completed
P4	5	1	0	4	9	10	4	5	Activate Windows Go to Settings to activate Windows.	Completed
P2	4	1	0	3	6	7	2	3	2	Completed

MLFQ:



Learning Outcomes

Through this project, I learned:

1. How CPU scheduling algorithms work.
2. Difference between preemptive and non-preemptive scheduling.
3. How to calculate scheduling metrics.
4. How to create GUI applications using PyQt6.
5. How to structure a Python project properly.
6. How to visualize process execution using a Gantt chart.
7. How to compare algorithms based on performance.

Conclusion

The Real-Time Interactive CPU Scheduling Simulator successfully simulates multiple CPU scheduling algorithms using a graphical user interface. It allows users to add processes, select algorithms, view Gantt charts, and analyze performance metrics.

The project helps in understanding how different scheduling algorithms behave under different workloads. The algorithm comparison and adaptive feedback features make the simulator more useful for learning and analysis.

Overall, this project improves understanding of CPU scheduling concepts and demonstrates how operating system algorithms can be simulated using Python and PyQ